Wei Ling Tan

Year 3, Computer Science, Nanyang Technological University (CCDS)
Sample CV for the URECAMatch demo. This person and the organisations marked fictional do not exist.

Summary

Student researcher in medical image analysis. I have trained, evaluated and deployed deep learning models for retinal fundus images, including diabetic retinopathy grading.

Education

Bachelor of Computing in Computer Science, NTU, 2024 to present
CGPA 2.90 out of 5.00. Grades fell in my first two years while I worked part time. Deep Learning: A-. Computer Vision: A.

Research and projects

Diabetic retinopathy grading on APTOS 2019 (course project, then independent)
- Trained EfficientNet-B3 and ResNet-50 in PyTorch on 3,662 fundus images to grade severity from 0 to 4.
- Reached a quadratic weighted kappa of 0.89 on a held-out split, up from a 0.78 baseline.
- Handled class imbalance with focal loss and weighted sampling; added Ben Graham preprocessing and test time augmentation.
- Used Grad-CAM to check that the model attends to microaneurysms and haemorrhages.

Referable retinopathy screening on EyePACS
- Built a binary screening model on 35,000 images with AUC 0.96, sensitivity 91% and specificity 90%.
- Measured the drop in accuracy on a Messidor-2 external validation set and reduced it with colour augmentation.

Lesion segmentation with U-Net on IDRiD
- Segmented hard exudates with a Dice score of 0.74 and wrote up the results as a four page workshop style report.

Experience

Machine Learning Intern, Lumen Retina Health (fictional), May to August 2026
- Shipped a fundus image quality filter to the clinic pilot, which cut ungradable images reaching review by 32%.
- Built the training pipeline with PyTorch, MLflow and Docker, and exported models to ONNX.
- Worked with two ophthalmologists to relabel 1,200 disputed images.

Teaching Assistant, Introduction to Data Science and AI, NTU, January to May 2026
- Ran weekly labs for 40 students on CNNs and model evaluation.

Skills

Deep learning: PyTorch, torchvision, transfer learning, Grad-CAM, ONNX. **Medical imaging:** fundus preprocessing, OpenCV, Albumentations. **Evaluation:** quadratic weighted kappa, AUC, sensitivity and specificity, external validation. **Tools:** Python, Git, Docker, MLflow, Linux, Slurm GPU clusters.

Awards

Top 5% of 2,900 teams in a public retinal imaging Kaggle competition. Best Project, CCDS Deep Learning module showcase.